

INTERNATIONAL STANDARD

**Industrial communication networks – Fieldbus specifications –
Part 3-17: Data-link layer service definition – Type 17 elements**



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IEC Central Office
3, rue de Varembe
CH-1211 Geneva 20
Switzerland
Email: inmail@iec.ch
Web: www.iec.ch

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INDUSTRIAL COMMUNICATION NETWORKS – FIELDBUS SPECIFICATIONS –

Part 3-17: Data-link layer service definition – Type 17 elements

FOREWORD

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NOTE Use of some of the associated protocol types is restricted by their intellectual-property-right holders. In all cases, the commitment to limited release of intellectual-property-rights made by the holders of those rights permits a particular data-link layer protocol type to be used with physical layer and application layer protocols in type combinations as specified explicitly in the IEC 61784 series. Use of the various protocol types in other combinations may require permission of their respective intellectual-property-right holders.

International Standard IEC 61158-3-17 has been prepared by subcommittee 65C: Industrial networks, of IEC technical committee 65: Industrial-process measurement, control and automation.

This first edition and its companion parts of the IEC 61158-3 subseries cancel and replace IEC 61158-3:2003. This edition of this part constitutes a technical addition. This part and its Type 17 companion parts also replace IEC/PAS 62405, published in 2005.

This edition includes the following significant changes with respect to the previous edition:

- a) deletion of the former Type 6 fieldbus, and the placeholder for a Type 5 fieldbus data-link layer, for lack of market relevance;
- b) addition of new types of fieldbuses;
- c) division of this part into multiple parts numbered 3-1, 3-2, ..., 3-19.

This edition of this part constitutes an editorial revision.

The text of this standard is based on the following documents:

FDIS	Report on voting
65C/473/FDIS	65C/484/RVD

Full information on the voting for the approval of this standard can be found in the report on voting indicated in the above table.

This publication has been drafted in accordance with ISO/IEC Directives, Part 2.

The committee has decided that the contents of this publication will remain unchanged until the maintenance result date indicated on the IEC web site under <http://webstore.iec.ch> in the data related to the specific publication. At this date, the publication will be:

- reconfirmed;
- withdrawn;
- replaced by a revised edition, or
- amended.

NOTE The revision of this standard will be synchronized with the other parts of the IEC 61158 series.

The list of all the parts of the IEC 61158 series, under the general title *Industrial communication networks – Fieldbus specifications*, can be found on the IEC web site.

INTRODUCTION

This part of IEC 61158 is one of a series produced to facilitate the interconnection of automation system components. It is related to other standards in the set as defined by the “three-layer” fieldbus reference model described in IEC/TR 61158-1.

Throughout the set of fieldbus standards, the term “service” refers to the abstract capability provided by one layer of the OSI Basic Reference Model to the layer immediately above. Thus, the data-link layer service defined in this standard is a conceptual architectural service, independent of administrative and implementation divisions.

INDUSTRIAL COMMUNICATION NETWORKS – FIELDBUS SPECIFICATIONS –

Part 3-17: Data-link layer service definition – Type 17 elements

1 Scope

This part of IEC 61158 provides common elements for basic time-critical messaging communications between devices in an automation environment. The term “time-critical” is used to represent the presence of a time-window, within which one or more specified actions are required to be completed with some defined level of certainty. Failure to complete specified actions within the time window risks failure of the applications requesting the actions, with attendant risk to equipment, plant and possibly human life.

This standard defines in an abstract way the externally visible service provided by the Type 17 fieldbus data-link Layer in terms of

- a) the primitive actions and events of the service;
- b) the parameters associated with each primitive action and event, and the form which they take; and
- c) the interrelationship between these actions and events, and their valid sequences.

The purpose of this standard is to define the services provided to

- the Type 17 fieldbus application layer at the boundary between the application and data-link layers of the fieldbus reference model, and
- systems management at the boundary between the data-link layer and systems management of the fieldbus reference model;
- specifications

The principal objective of this standard is to specify the characteristics of conceptual data-link layer services suitable for time-critical communications, and thus supplement the OSI Basic Reference Model in guiding the development of data-link protocols for time-critical communications. A secondary objective is to provide migration paths from previously-existing industrial communications protocols.

This specification may be used as the basis for formal DL-Programming-Interfaces. Nevertheless, it is not a formal programming interface, and any such interface will need to address implementation issues not covered by this specification, including

- a) the sizes and octet ordering of various multi-octet service parameters, and
 - b) the correlation of paired request and confirm, or indication and response, primitives.
- Conformance

This standard does not specify individual implementations or products, nor does it constrain the implementations of data-link entities within industrial automation systems.

There is no conformance of equipment to this data-link layer service definition standard. Instead, conformance is achieved through implementation of the corresponding data-link protocol that fulfills the Type 17 data-link layer services defined in this standard.

2 Normative reference

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For all other undated references, the latest edition of the referenced document (including any amendments) applies.

ISO/IEC 7498-1, *Information technology – Open Systems Interconnection — Basic Reference Model: The Basic Model*

ISO/IEC 7498-3, *Information technology – Open Systems Interconnection — Basic Reference Model: Naming and addressing*

ISO/IEC 10731:1994, *Information technology – Open Systems Interconnection – Basic Reference Model – Conventions for the definition of OSI services*

ISO/IEC 8802-3, *Information technology – Telecommunications and information exchange between systems – Local and metropolitan area networks – Specific requirements – Part 3: Carrier sense multiple access with collision detection (CSMA/CD) access method and physical layer specifications*

Internet Engineering Task Force (IETF), *Request for Comments (RFC)*:

RFC 826 *Ethernet Address Resolution Protocol*
(available at <<http://www.ietf.org/rfc/rfc0826.txt>>)

3 Definitions

For the purposes of this document, the following terms and definitions apply.

3.1 Terms and definitions

3.1.1 ISO/IEC 10731 terms

- a) (N)-connection
- b) (N)-entity
- c) (N)-layer
- d) (N)-service
- e) (N)-service-access-point
- f) confirm (primitive)
- g) deliver (primitive)
- h) indication (primitive)
- i) request (primitive)
- j) response (primitive)

3.1.2 Other terms and definitions

3.1.2.1

bridge

intermediate equipment that connects two or more segments using a data-link layer relay function

3.1.2.2

domain

part of the RTE network consisting of one or two subnetwork(s)

NOTE Two subnetworks are required to compose a dual-redundant RTE network, and each end node in the domain is connected to both of the subnetworks.

3.1.2.3**domain master**

station which performs diagnosis of routes to all other domains, distribution of network time to nodes inside the domain, acquisition of absolute time from the network time master and notification of status of the domain

3.1.2.4**domain number**

numeric identifier which indicates a domain

3.1.2.5**external bridge**

bridge to which neither internal bridges nor RTE stations are connected directly

3.1.2.6**interface port**

physical connection point of an end node, which has an independent DL-address

3.1.2.7**internal bridge**

bridge to which no routers, external bridges or nodes non-compliant with this specification are connected directly

3.1.2.8**junction bridge**

bridge to which at least one router, external bridge or node non-compliant with this specification, and to which at least one internal bridge or RTE station is connected

3.1.2.9**link**

physical communication channel between two nodes

3.1.2.10**network time master**

station which distributes network time to domain masters

3.1.2.11**non-redundant interface node**

node which has a single interface port

3.1.2.12**non-redundant station**

station that consists of a single end node

NOTE "non-redundant station" is synonymous with "end node".

3.1.2.13**path**

logical communication channel between two nodes, which consists of one or two link(s)

3.1.2.14**redundant interface node**

node with two interface ports one of which is connected to a primary network, while the other is connected to a secondary network

3.1.2.15**redundant station**

station that consists of a pair of end nodes

NOTE Each end node of a redundant station has the same station number, but has a different DL-address.

3.1.2.16

route

logical communication channel between two communication end nodes

3.1.2.17

router

intermediate equipment that connects two or more subnetworks using a network layer relay function

3.1.2.18

RTE station

station with real-time capability

3.1.2.19

segment

communication channel that connects two nodes directly without intervening bridges

3.1.2.20

station

end node or a pair of end nodes that perform a specific application function

3.1.2.21

station number

numeric identifier which indicates a RTE station

3.1.2.22

subnetwork

part of a network that does not contain any routers. A subnetwork consists of end nodes, bridges and segments

NOTE Every end node included in a subnetwork has the same IP network address.

3.2 Abbreviations and symbols

3.2.1 ISO/IEC 10731 abbreviations

OSI	Open Systems Interconnection
------------	------------------------------

3.2.2 Other abbreviations and symbols

ASS	Acknowledged sequence of unitdata transfer service
AUS	Acknowledged unitdata transfer service
cnf	Confirmation primitive
DL-	Data-link layer (as a prefix)
DLE	DL-entity (the local active instance of the data-link layer)
DLL	DL-layer
DLM	DL-management
DLMS	DL-management service
DLPDU	DL-protocol-data-unit
DLS	DL-service
DLSAP	DL-service-access-point
DLSDU	DL-service-data-unit
FIFO	First-in first-out (queuing method)
ind	Indication primitive

IP	Internet protocol
ISO	International Organization for Standardization
PDU	Protocol data unit
MSS	Multipoint sequence of unitdata transfer service
MUS	Multipoint unitdata transfer service
QoS	Quality of service
req	Request primitive
rsp	Response primitive
SAP	Service access point
SDU	Service data unit
ToS	Type of service
UUS	Unacknowledged unitdata transfer service

3.3 Conventions

This standard uses the descriptive conventions given in ISO/IEC 10731.

The service model, service primitives, and time-sequence diagrams used are entirely abstract descriptions; they do not represent a specification for implementation.

Service primitives, used to represent service user/service provider interactions (see ISO/IEC 10731), convey parameters that indicate information available in the user/provider interaction.

This standard uses a tabular format to describe the component parameters of the DLS primitives. The parameters that apply to each group of DLS primitives are set out in tables throughout the remainder of this standard. Each table consists of up to six columns, containing the name of the service parameter, and a column each for those primitives and parameter-transfer directions used by the DLS:

- the request primitive's input parameters;
- the request primitive's output parameters;
- the indication primitive's output parameters;
- the response primitive's input parameters; and
- the confirm primitive's output parameters.

NOTE The request, indication, response and confirm primitives are also known as requestor.submit, acceptor.deliver, acceptor.submit, and requestor.deliver primitives, respectively (see ISO/IEC 10731).

One parameter (or part of it) is listed in each row of each table. Under the appropriate service primitive columns, a code is used to specify the type of usage of the parameter on the primitive and parameter direction specified in the column:

- | | |
|----------|---|
| M | — parameter is mandatory for the primitive. |
| U | — parameter is a User option, and may or may not be provided depending on the dynamic usage of the DLS-user. When not provided, a default value for the parameter is assumed. |
| C | — parameter is conditional upon other parameters or upon the environment of the DLS-user. |
| (blank) | — parameter is never present. |

Some entries are further qualified by items in brackets. These may be

- a) a parameter-specific constraint
 - (=) indicates that the parameter is semantically equivalent to the parameter in the service primitive to its immediate left in the table;
- b) an indication that some note applies to the entry
 - (n) indicates that the following note n contains additional information pertaining to the parameter and its use.

In any particular interface, not all parameters need be explicitly stated. Some may be implicitly associated with the DLSAP at which the primitive is issued.

In the diagrams which illustrate these interfaces, dashed lines indicate cause-and-effect or time-sequence relationships, and wavy lines indicate that events are roughly contemporaneous.

4 Overview of the data-link layer service

4.1 General

The data-link service (DLS) provides transparent and reliable data transfer between DLS-users. It makes the way that supporting communication resources are utilized invisible to DLS-users.

In particular, the DLS provides the following.

- a) Independence from the underlying Physical Layer. The DLS relieves DLS-users from all direct concerns regarding which configuration is available (for example, direct connection, or indirect connection through one or more bridges) and which physical facilities are used (for example, which of a set of diverse physical paths).
- b) Transparency of transferred information. The DLS provides the transparent transfer of DLS-user-data. It does not restrict the content, format or coding of the information, nor does it ever need to interpret the structure or meaning of that information. It may, however, restrict the amount of information that can be transferred as an indivisible unit.
- c) Reliable data transfer. The DLS relieves the DLS-user from concerns regarding insertion or corruption of data, or, if requested, loss, duplication or misordering of data, which can occur. In some cases of unrecovered errors in the data-link layer, duplication or loss of DLSUs can occur. In cases where protection against misordering of data is not employed, misordering can occur.
- d) Quality of Service (QoS) selection. The DLS provides DLS-users with a means to request and to agree upon a quality of service for the data transfer. QoS is specified by means of QoS parameters representing aspects such as mode of operation, transit delay, accuracy, reliability, security and functional safety.
- e) Addressing. The DLS allows the DLS-user to identify itself and to specify the DLSAPs to/from which data are to be transferred.
- f) Scheduling. The DLS allows the set of DLS-users to provide some guidance on internal scheduling of the distributed DLS-provider. This guidance supports the time-critical aspects of the DLS, by permitting the DLS-user some degree of management over when opportunities for communication will be granted to various DLEs for various DLSAP-addresses.
- g) Common time sense. The DLS can provide the DLS-user with a sense of time that is common to all DLS-users on the network.
- h) Queues. The DLS can provide the sending or receiving DLS-user with a FIFO queue, where each queue item can hold a single DLSU.

4.2 Overview of network structure

Although the DLS conforms formally to the “three-layer” Fieldbus Reference Model, it actually utilizes the transport layer service and the network layer service in addition to the data-link layer service of the OSI Basic Reference Model. The DLS of this specification is actually a transport layer service in terms of the OSI Basic Reference Model. Thus the network may consist of one or more subnetworks interconnected to each other by the network layer relay entities, known as routers.

A network may be a redundant structure. A redundant network consists of two independent networks making dual-redundancy; they are referred to as the primary network and the secondary network. Consequently, dual-redundant independent logical communication channels between two communication end nodes can be implemented. This logical channel is called a route.

A pair of subnetworks comprising a dual-redundant network is called a domain.

A subnetwork consists of one or more segments interconnected by DL-relay entities, known as bridges. The topology of a subnetwork may be a tree, a ring or a mesh consisting of segments interconnected by bridges.

A segment consists of one or more DLEs, all of which are connected directly (i.e., without intervening DL-relay entities) to a single shared logical communication channel, which is called a link.

A path (logical communication channel) consists of one or two physically independent and logically parallel real communication channels, which are called links.

4.3 Overview of addressing

domain number

numeric identifier that indicates a domain. Two subnetworks comprising a dual-redundant domain have an identical domain number.

station number

numeric identifier that indicates a RTE station. Two end nodes comprising a dual-redundant station have an identical station number.

TSAP address

DL-entity actually provides transport layer services, so DLS is provided at TSAPs. TSAP is identified by a set of TSAP-address (IP-address) and TSAP-identifier (UDP port number)

IP address

unique address for each end node. An IP address consists of a network address portion and a host address portion. The network address is assigned according to the domain number, while the host address is assigned based on the station number. Each end node of a dual-redundant station has a different host address

MAC address

MAC address is a unique address for an end node defined in ISO/IEC 8802-3. The destination MAC address is resolved by the mechanism defined in RFC 826 from the destination IP address

4.4 Types of data-link service

There are three types of DLS as follows:

- a) a DLSAP management service;
- b) a connectionless-mode data transfer service;
- c) a DL-management service.

5 DLSAP management service

5.1 Overview

This clause provides a conceptual definition of the services provided by the DLS-provider to the DLS-user(s). This clause does not constrain the actual implementations of the interactions at the DLS-provider to the DLS-user interface.

5.2 Facilities of the DLSAP management service

The DLS provides the following facilities to the DLS-user:

- a) a means for creating and deleting a FIFO queue of specified depth;
- b) a means for assigning a DLSAP-address to the DLSAP;
- c) a means for binding previously created FIFO queues to each potential direction of connectionless data transfer at the specified DLSAP;
- d) a means for specifying QoS parameters of the specified DLSAP;
- e) a means for releasing resources used previously for the DLSAP.

5.3 Model of the DLSAP management service

This standard uses the abstract model for a layer service defined in ISO/IEC 10731, Clause 5. The model defines interactions between the DLS-user and the DLS-provider that take place at a DLSAP. Information is passed between the DLS-user and the DLS-provider by DLS primitives that convey parameters.

The DLSAP management primitives are used to provide a local service between a DLS-user and the local DLE. Remote DLEs and remote DLS-users are not involved direct, so there is no need for the other primitives of ISO/IEC 10731. Therefore the DLSAP management services are provided by request primitives with input and output parameters.

5.4 Sequence of primitives at one DLSAP

Table 1 is a summary of the DLSAP management primitives and parameters. The major sequence of primitives at a single DLE is shown in Figure 1.

Table 1 – Summary of DLSAP management primitives and parameters

Service	Primitive	Parameter
queue creation	DL-CREATE request	(<i>in</i> Queue DLS-user-identifier, Maximum DLSDU size, Maximum queue depth, <i>out</i> Status, Queue DL-identifier)
queue deletion	DL-DELETE request	(<i>in</i> Queue DL-identifier, <i>out</i> Status)
DLSAP activation	DL-BIND request	(<i>in</i> DLSAP-address DLS-user-identifier, Sending queue DL-identifier Receiving queue DL-identifier, QoS parameters, <i>out</i> Status, DLSAP-address DL-identifier)
DLSAP deactivation	DL-UNBIND request	(<i>in</i> DLSAP-address DL-identifier <i>out</i> Status)

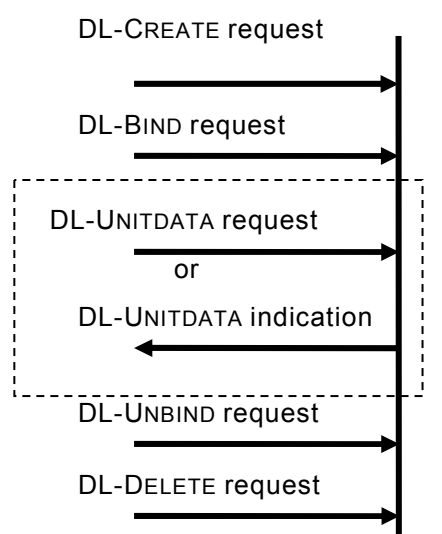


Figure 1 – Sequence of primitives for the DLSAP management DLS

5.5 Create

5.5.1 Function

The create queue DLS primitive may be used to create a limited-depth FIFO queue for later constrained association with a DLSAP. The resulting FIFO queue initially will be empty.

5.5.2 Types of parameter

Table 2 lists the primitive and parameters of the create queue DLS.

Table 2 – DLSAP-management CREATE primitive and parameters

DL-CREATE Parameter name	Request	
	input	output
Queue DLS-user-identifier	M	
Maximum DLSDU size	M	
Maximum queue depth	M	
Status		M
Queue DL-identifier		C

5.5.2.1 Queue DLS-user-identifier

This parameter specifies a means of referring to the queue in output parameters of other local DLS primitives that convey the name of the queue from the local DLE to the local DLS-user.

The naming-domain of the queue DLS-user-identifier is the DLS-user's local-view.

5.5.2.2 Maximum DLSDU size

This parameter specifies an upper bound on the size (in octets) of DLSDUs that can be put into the queue.

5.5.2.3 Maximum queue depth

This parameter specifies the maximum number of items in the associated queue.

5.5.2.4 Status

This parameter allows the DLS-user to determine whether the requested DLS was provided successfully, or failed for the reason specified. The possible value conveyed in this parameter is as follows:

- a) "success";
- b) "failure — insufficient resources";
- c) "failure — parameter violates management constraint";
- d) "failure — number of requests violates management constraint"; or
- e) "failure — reason unspecified".

NOTE Addition to, or refinement of, this list of values to convey more specific diagnostic and management information is permitted.

5.5.2.5 Queue DL-identifier

This parameter is present when the Status parameter indicates that the DL-CREATE request primitive was successful. The queue DL-identifier parameter gives the local DLS-user a means of referring to the queue in input parameters of other local DLS primitives that convey the name of the queue from the local DLS-user to the local DLE.

5.6 Delete

5.6.1 Function

The delete queue DLS primitive may be used to delete a queue created by an earlier create queue DLS primitive.

5.6.2 Types of parameter

Table 3 lists the primitive and parameters of the delete queue DLS.

Table 3 – DLSAP-management DELETE primitive and parameters

DL-DELETE Parameter name	Request	
	input	output
Queue DL-identifier	M	
Status		M

5.6.2.1 Queue DL-identifier

This parameter specifies the local queue to be deleted. Its value is the queue DL-identifier returned by a successful prior DL-CREATE request primitive. The DLS-provider will release the local DL-identifier and associated DLS-provider resources.

The DLS-user may not delete a queue that is still associated with a DLSAP.

5.6.2.2 Status

This parameter allows the DLS-user to determine whether the requested DLS was provided successfully or failed for the reason specified. The value conveyed in this parameter is as follows:

- a) "success";
- b) "failure — resource in use"; or
- c) "failure — reason unspecified".

NOTE Addition to, or refinement of, this list of values to convey more specific diagnostic and management information is permitted.

5.7 Bind

5.7.1 Function

The bind DLSAP-address DLS primitive is used

- a) to associate a DLSAP-address with the DLSAP at which the primitive is invoked;
- b) to associate previously created limited depth FIFO queues with the various priorities and directions of potential data transfer at the specified DLSAP-address; and
- c) to specify values for some Quality of Service (QoS) attributes for connectionless data transfer services using the specified DLSAP-address.

5.7.2 Types of parameter

Table 4 lists the primitive and parameters of the Bind DLSAP-address DLS.

Table 4 – DLSAP management BIND primitive and parameters

DL-BIND Parameter name	Request	
	input	output
DLSAP-address DLS-user-identifier	M	
Sending queue DL-identifier	M	
Receiving queue DL-identifier	M	
QoS parameters		
DL Service subtype	M	
DLL maximum confirm delay	U	
DLL priority	U	
Authentication level	U	
Maximum residual error rate	U	
Transmission Timing Window	U	
Status		M
DLSAP-address DL-identifier		C

5.7.2.1 DLSAP-address DLS-user-identifier

This parameter specifies a means of referring to the DLSAP-address in output parameters of other local DLS primitives that convey the name of the DLSAP-address from the local DLE to the local DLS-user.

5.7.2.2 Sending queue DL-identifier

This parameter specifies the local DL-identifier for sending, which has been returned by a successful prior DL-CREATE request primitive that created a queue, which has not yet been deleted.

5.7.2.3 Receiving queue DL-identifier

This parameter specifies the local DL-identifier for receiving, which has been returned by a successful prior DL-CREATE request primitive that created a queue, which has not yet been deleted.

5.7.2.4 QoS parameters

The DLS-user may specify values for some of the QoS parameters that apply to connectionless data transmission. If a value is not specified, the default value set by DL-management is applied for the DLSAP

5.7.2.5 DL Service subtype

This parameter specifies service subtype of the DLSAP, and the value conveyed in this parameter is as follows.

- a) "UUS" -- Unacknowledged Unitdata transfer Service
- b) "AUS" -- Acknowledged Unitdata transfer Service
- c) "ASS" -- Acknowledged Sequence of unitdata transfer Service
- d) "MUS" -- Multipoint Unitdata transfer Service
- e) "MSS" -- Multipoint Sequence of unitdata transfer Service

The function of each service subtype is specified in Clause 3.

5.7.2.6 DLL maximum confirm delay

This parameter specifies the value of the maximum confirm delay.

5.7.2.7 DLL priority

This parameter specifies the value of the DLL priority.

5.7.2.8 Authentication level

This parameter specifies the level of the DLL authentication.

5.7.2.9 Maximum residual error rate

This parameter specifies a value of the permissible residual error rate.

5.7.2.10 Transmission Timing Window

This parameter specifies the timing window within the macro cycle. It consists of the starting time in the macro cycle and the time duration when DLE is permitted to transmit DL-UNITDATA DLPDUs.

5.7.2.11 Status

This parameter allows the DLS-user to determine whether the requested DLS was provided successfully or failed for the reason specified. The possible value conveyed in this parameter is as follows:

- a) "success";
- b) "failure — insufficient resources";
- c) "failure — DLSAP-address invalid or unavailable";
- d) "failure — invalid queue binding";
- e) "failure — requested QoS attribute is not available"; or
- f) "failure — reason unspecified".

NOTE Addition to, or refinement of, this list of values to convey more specific diagnostic and management information is permitted.

5.7.2.12 DLSAP-address DL-identifier

The DLSAP-address DL-identifier parameter is present when the status parameter indicates that the DL-BIND request primitive was successful. The DLSAP-address DL-identifier parameter gives the local DLS-user a means of referring to the DLSAP-address in input parameters of other local DLS primitives that convey the name of the DLSAP-address from the local DLS-user to the local DLE.

The naming-domain of the DLSAP-address DL-identifier is the DL-local-view.

5.8 Unbind

5.8.1 Function

The unbind DLSAP-address DLS primitive is used to unbind a DLSAP-address from the invoking DLSAP. Any queues that were bound explicitly to the DLSAP-address are also unbound from that DLSAP-address at the same time.

5.8.2 Types of parameter

Table 5 lists the primitive and parameter of the unbind DLSAP-address DLS.

Table 5 – DLSAP management UNBIND primitive and parameter

DL-UNBIND Parameter name	Request	
	input	output
DLSAP-address DL-identifier	M	
Status		M

5.8.2.1 DLSAP-address DL-identifier

This parameter specifies the local DL-identifier returned by a successful prior DL-BIND request primitive. The DLS-provider unbinds the local DL-identifier, its associated DLSAP-address, and any associated queues from the invoking DLSAP and terminates all outstanding DL-UNITDATA requests associated with that DLSAP-address.

5.8.2.2 Status

This parameter allows the DLS-user to determine whether the requested DLS was provided successfully, or failed for the reason specified. The possible value conveyed in this parameter is as follows:

- a) "success";
- b) "failure – reason unspecified".

NOTE Addition to, or refinement of, this list of values to convey more specific diagnostic and management information is permitted.

6 Connectionless-mode data-link service

6.1 Overview

This clause provides a conceptual definition of the services provided by the DLS-provider to the DLS-user(s). This clause does not constrain the actual implementations of the interactions at the DLS-provider to DLS-user interface.

The DLS provides two types of Data Transfer Services:

- a) point-to-point Data Transfer Services;
- b) multipoint Data Transfer Services.

There are three types of point-to-point Data Transfer Services:

- a) unacknowledged unitdata transfer service (UUS);
- b) acknowledged unitdata transfer service (AUS);
- c) acknowledged sequence of unitdata transfer service (ASS).

There are two types of multipoint Data Transfer Service:

- a) multipoint unitdata transfer service (MUS);
- b) multipoint Sequence of unitdata transfer service (MSS).

6.2 Facilities of the connectionless-mode data-link service

The DLS provides the following facilities to DLS-user:

- a) a means of transferring DLSDUs with limited length from one source DLSAP to a destination DLSAP or a group of DLSAPs, without establishing or later releasing a data-link connection. The transfer of DLSDUs is transparent, which means that the boundaries of DLSDUs and the contents of DLSDUs are preserved unchanged by the DLS, and there are no constraints on the DLSDU content (other than limited length) imposed by the DLS provider. QoS for this transmission may be selected by the sending DLS-user;
- b) a means by which the status of delivery to that destination DLSAP can be returned to the source DLSAP.

6.3 Model of the data-link service

6.3.1 General

This clause uses the abstract model for a layer service defined in ISO/IEC 10731, Clause 5. The model defines local interactions between the DLS-user and the DLS-provider that take place at a DLSAP. Information is passed between the DLS-user and the DLS-provider by DLS primitives that convey parameters.

A defining characteristic of data-link connectionless-mode unitdata transmission is basically the independent nature of each invocation of the DLS.

6.3.2 DLS-instance identification

A local identification mechanism is provided for each use of the DLS which needs to correlate a confirmation or subsequent cancellation request with its associated request.

DL-connectionless-mode service primitives may be used to transmit independent DLSDUs from one DLSAP to another DLSAP or other DLSAPs. Each DLSDU is transmitted in a single DLPDU. The DLSDU is independent in the sense that it bears no relationship to any other DLSDU transmitted through an invocation of the DLS. The DLSDU is self-contained in that all the information required to deliver the DLSDU is presented to the DLS provider, together with the user data to be transmitted, in a single service access.

6.3.3 Unacknowledged unitdata transfer service (UUS)

This service permits a local DLS-user to transfer a DLSDU to a single remote end node. The local DLS-user receives a confirmation acknowledging the completion of the transfer, but not whether the DLSDU was duly received. At the remote end node this DLSDU is delivered to a single local DLS-user, if the respective DLPDU is received error-free. There is no confirmation to the sending DLS-user that such an intended delivery has taken place.

6.3.4 Acknowledged unitdata transfer service (AUS)

This service permits the local DLS-user to send a DLSDU to a single remote end node. At the remote end node, the DLSDU is delivered by the remote DLE to its local DLS-user, if the respective DLPDU is transferred error-free. The originating local DLS-user receives a confirmation that indicates whether the DLSDU was received by the remote DLS-user. If an error occurred during the transfer, the originating DLE retransmits the DLSDU appropriately. If the specified number of retransmissions fail, a confirmation with error status will be issued to the local DLS-user.

6.3.5 Acknowledged sequence of unitdata transfer service (ASS)

This service permits the local DLS-user to send a sequence of DLSDUs to a single remote end node. At the remote end node, the DLSDU are delivered in the original order requested by the remote DLE to its local DLS-user. The originating local DLS-user receives confirmation of completion of the transmission of each DLSDU, but does not receive confirmation of receipt of the DLSDU by the remote DLS-user. If an error occurred during transfer, the originating DLE retransmits the DLSDUs appropriately. A sequence of DLSDUs is formed from the requested DLSDUs by originating DLE according to a preconfigured rule.

6.3.6 Multipoint unitdata transfer service (MUS)

This service permits a local DLS-user to transfer a DLSDU to the preconfigured group of remote end nodes at the same time. The local DLS-user receives confirmation of completion of the transmission, but does not receive confirmation of receipt of the DLSDU by the remote DLS-user. At each addressed remote end node, this DLSDU is delivered to the remote DLS-user, if the respective DLPDU is received error-free. There is no confirmation to the sending DLS-user that such an intended delivery has taken place.

6.3.7 Multipoint sequence of unitdata transfer service (MSS)

This service permits a local DLS-user to transfer a DLSDU to a preconfigured group of remote end nodes at the same time. The local DLS-user receives confirmation of completion of the transmission, but does not receive confirmation of receipt of the DLSDU by the remote DLS-user. At each remote addressed end node, the DLSDU are delivered to the remote DLS-user in the original requested order.

6.4 Quality of service

A DLS-user may select many aspects of the various data-link services. The term “Quality of Service” (QoS) refers to certain characteristics of a connectionless-mode data transmission as observed between the DLSAPs. QoS describes aspects of data transmission that are with a relation solely to the DLS-provider. QoS can only be determined properly when DLS-user behaviour does not constrain or impede performance of the DLS.

The QoS attributes of the DL-data transfer services apply conceptually to the connectionless operation. The DLS-user may specify values for these attributes when binding a DLSAP-address to the DLS-user’s DLSAP; any unspecified attributes will assume the default values set by DL-management.

6.4.1 Data delivery features

The data delivery feature depends on the type of service, as specified in Table 6.

Table 6 – Data delivery features of each type of service

Service Type	Action on lost DLSDUs	Action on duplicated DLSDUs	Action on misordered DLSDUs	Action on lack of receive buffer
UUS	not detected	delivered as received	delivered as received	discarded
AUS	recovered	discarded	delivered as received	wait & retransmitted
ASS	recovered conditionally	discarded	delivered in order	wait & retransmitted
MUS	not detected	delivered as received	delivered as received	discarded
MSS	detected	discarded	delivered as received	discarded

6.4.2 DLL priority

All connectionless data transfer requests specify an associated DLL priority used in scheduling DLL data transfer services.

The choice of sending priority implicitly selects the timing characteristics of the DLS provider execution of the transmission. The following four alternative priorities are available:

- a) URGENT,
- b) HIGH,
- c) NORMAL, and
- d) TIME-AVAILABLE.

The DLSDU requested to transmit with higher priority will be transmitted earlier than those with lower priority.

6.4.3 DLL maximum confirm delay

Each connectionless service request specifies an upper bound on the maximum time duration permitted for the completion of each related instance of DLS primitives:

- a) the maximum time for completion of a related sequence of locally confirmed DL-UNITDATA primitives; and
- b) the maximum time for completion of a related sequence of remotely confirmed DL-UNITDATA primitives.

Each parameter has the value **UNLIMITED** or specifies an upper bound in units of 1 ms, from 1 ms to 60 s inclusive.

6.4.4 Authentication level

This attribute specifies the level of authentication for data transfer. The following four alternative levels are available:

- a) “no authentication”;
- b) “use 64-bit key code”;
- c) “use 128-bit key code”;
- d) “use 256-bit key code”.

6.4.5 Maximum residual error rate

This parameter specifies the upper bound on acceptable residual error rate of the underlying layer service.

The DLL monitors the bit error rate of the underlying layer service continuously. Under conditions where the residual error rate is higher than the maximum residual error rate, the requested DL-UNITDATA request is completed with error. The residual error rate is calculated from the bit error rate and the error detection performance.

This feature allows the DLS-user to switch the sending interface to prevent unexpected loss of data integrity caused by the underlying service.

6.4.6 Transmission Timing Window

This parameter specifies the timing window within the macro cycle. It consists of the starting time in the macro cycle and the time duration when DLE is permitted to transmit DL-UNITDATA DLPDUs.

6.5 Sequence of primitives

This subclause defines the constraints on the sequence in which the primitives defined may occur. The constraints determine the order in which primitives occur, but do not fully specify when they may occur. Other aspects of actual system operation, such as PhL problems affecting messages in transit, will affect the ability of a DLS-user or a DLS provider to issue a primitive at any particular time.

The connectionless-mode primitives and their parameters are summarized in Table 7.

Table 7 – Summary of DL-connectionless-mode primitives and parameters

Service	Primitive	Parameter
Data transfer	DL-UNITDATA request	(<i>in</i> Called DLSAP-address, Calling DLSAP-address DL-identifier, limited DLS-user-data)
	DL-UNITDATA indication	(<i>out</i> Called DLSAP-address DLS-user-identifier, Calling DLSAP-address, Queue DLS-user-identifier, limited DLS-user-data)
	DL-UNITDATA confirm	(<i>out</i> Transfer-Status)

6.5.1 Relation of primitives at the endpoints of connectionless service

A primitive issued at one DLSAP will have consequences at one or more other DLSAPs. The relations of primitives are summarized in the diagrams of Figure 2.

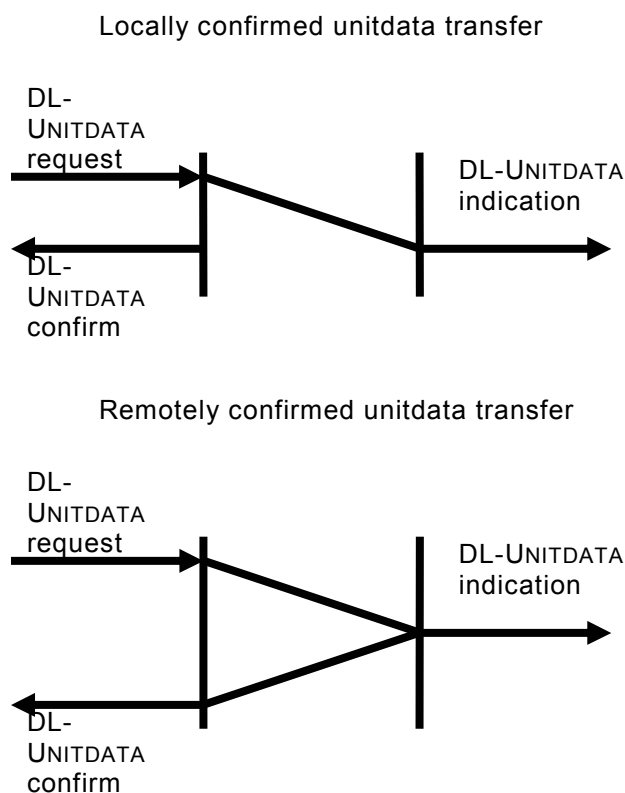


Figure 2 – Summary of DL-connectionless-mode service primitive time-sequence diagrams

6.5.2 Sequence of primitives at one DLSAP

The possible overall sequences of primitives at a DLSAP are defined in the state transition diagram, Figure 3. The state transition diagram describes the allowable sequences of service primitives but does not impose any requirements or constraints on the internal organization of any implementation of the service.

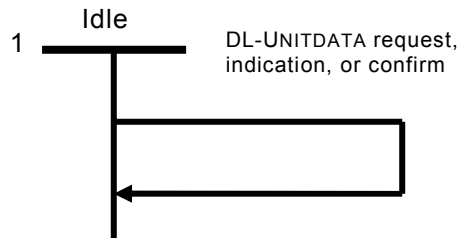


Figure 3 – State transition diagram for sequences of connectionless-mode primitives at one DLSAP

6.6 Connectionless-mode function

DL-connectionless-mode unitdata transmission service primitives may be used to transmit independent DLSDUs from one DLSAP to another. Each DLSDU is transmitted in a single DLPDU. The DLSDU is independent in the sense that it bears no relationship to any other DLSDU transmitted through an invocation of the DLS. The DLSDU is self-contained in that all the information required to deliver the DLSDU is presented to the DLS-provider, together with the user data to be transmitted, in a single service access.

A DLSDU transmitted using DL-connectionless-mode unitdata transmission services is not considered by the DLS-provider to be related in any way to any other DLSDU. The DLS maintains the integrity of individual DLSDUs, and delivers them to the receiving DLS-user in the order in which they are presented by the sending DLS-user if it is determined by the QoS parameter.

For the remote-confirmed service, the DLS provides the means by which the receiving DLS-user may control the rate at which the sending DLS-user may send DLSDUs.

6.7 Types of primitives and parameters

Table 8 lists the primitive and parameters of the DL-connectionless-mode UNITDATA transfer DLS.

Table 8 – DL-connectionless-mode UNITDATA transfer primitives and parameters

DL-UNITDATA Parameter name	Request	Indication	Confirm
	input	output	output
Called address	M		
Calling address		M	
Queue DLS-user-identifier		C	
DLS-user-data	M	C(=)	
Status			M

6.7.1 Called address

This parameter conveys an address identifying the remote DLSAP(s) with which the DLS is to be provided. It is a DLSAP-address in the request primitive, but takes the form of a local

DLSAP-address and DLS-user-identifier in the indication primitive(s). It may be a DLSAP-address or a group DL-address.

6.7.2 Calling address

This parameter conveys an address of the local DLSAP from which the DLS has been requested. It is a DLSAP-address in the indication primitive, but takes the form of a local DLSAP-address and DL-identifier in the request primitive.

6.7.3 Queue DLS-user-identifier

This parameter has the same value as a Queue DLS-user-identifier parameter from prior DL-CREATE primitive.

6.7.4 DLS-user-data

This parameter allows the data transfer between DLS-users without alteration by the DLS-provider. The initiating DLS-user may transmit any integral number of octets greater than zero up to the limit number determined by the Maximum DLSDU size parameter from a prior DL-CREATE primitive.

6.7.5 Status

This parameter allows the DLS-user to determine whether the requested service was provided successfully, or failed for the reason specified. The possible value conveyed in this parameter is as follows:

- a) "success";
- b) "failure — sending queue full";
- c) "failure — no requesting DLS-user data specified";
- d) "failure — requested QoS unavailable";
- e) "failure — calling DLSAP DL-identifier is invalid";
- f) "failure — terminated by UNBIND of source DLSAP-address";
- g) "failure — resource limitation in responder";
- h) "failure — fault in responder";
- i) "failure — high bit error rate";
- j) "failure — timeout before transmission";
- k) "failure — timeout after transmission, before acknowledgment";
- l) "failure — sequence number incorrect"; or
- m) "failure — reason unspecified".

NOTE Addition to, or refinement of, this list of values to convey more specific diagnostic and management information is permitted.

7 DL-management Service

7.1 Scope and inheritance

This clause defines the form of DL-management services for protocols that implement the DLS. Only the form is specified, as the specifics of permitted parameters are dependent on the protocol that implements these services.

The noteworthy difference between this and the previous clauses is the intended class of users; this clause is intended for use by a management client, while the previous clauses provide services to any client.

7.2 Facilities of the DL-management service

DL-management facilities provide a means for

- a) writing DLE configuration parameters;
- b) reading DLE configuration parameters, operational parameters and statistics;
- c) commanding major DLE actions; and
- d) receiving notification of significant DLE events.

Together, these facilities constitute the DL-management-service (DLMS).

7.3 Model of the DL-management service

This clause uses the abstract model for a layer service defined in ISO/IEC 10731, Clause 5. The model defines local interactions between the DLMS-user and the DLMS-provider. Information is passed between the DLMS-user and the DLMS-provider by DLMS primitives that convey parameters.

7.4 Constraints on sequence of primitives

This subclause defines the constraints on the sequence in which the primitives defined may occur. The constraints determine the order in which primitives occur, but do not fully specify when they may occur.

The DL-management primitives and their parameters are summarized in Table 9. The only time-sequence of primitives is shown in Figure 4.

Table 9 – Summary of DL-management primitives and parameters

Service	Primitive	Parameter
writing managed objects	DLM-SET request	(<i>in</i> DLM-object-identifier, Desired-value, <i>out</i> Status)
reading managed objects	DLM-GET request	(<i>in</i> DLM-object-identifier, <i>out</i> Status Current-value)
commanding actions	DLM-ACTION request	(<i>in</i> Desired-action, Action-qualifiers)
	DLM-ACTION confirm	<i>out</i> Status Additional-information)
notification of events	DLM-EVENT indication	(<i>out</i> DLM-event-identifier, Additional-information)

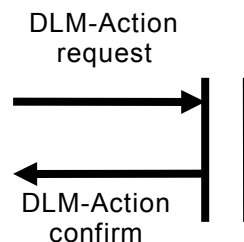


Figure 4 – Sequence of primitives for the DLM action service

7.5 Set

This primitive may be used to set (write) the value of a DLE configuration parameter.

7.5.1 Types of parameter

Table 10 lists the primitive and parameters of the SET DLMS.

Table 10 – DLM-SET primitive and parameters

DLM-SET Parameter name	Request	
	input	output
DLM-object-identifier	M	
Desired-value	M	
Status		M

7.5.1.1 DLM-object-identifier

This parameter specifies the primitive or composite object within the DLE the value of which is to be altered. The naming-domain of the DLM-object-identifier is the DLM-local-view.

7.5.1.2 Desired-value

This parameter specifies the desired value for the DLM-object specified by the associated DLM-object-identifier. Its type is identical to that of the specified DLM-object.

7.5.1.3 Status

This parameter allows the DLMS-user to determine whether the requested DLMS was provided successfully, or failed for the reason specified. The possible value conveyed in this parameter is as follows:

- a) "success";
- b) "failure — DLM-object-identifier is unknown";
- c) "failure — desired-value is not permitted"; or
- d) "failure — reason unspecified".

7.6 Get

7.6.1 Function

This primitive may be used to get (read) the value of a DLE configuration parameter, operational parameter or statistic.

7.6.2 Types of parameter

Table 11 lists the primitive and parameters of the GET DLMS.

Table 11 – DLM-GET primitive and parameters

DLM-GET Parameter name	Request	
	input	output
DLM-object-identifier	M	
Status		M
Current-value		C

7.6.2.1 DLM-object-identifier

This parameter specifies the primitive or composite object within the DLE whose value is being requested. The naming-domain of the DLM-object-identifier is the DLM-local-view.

7.6.2.2 Status

This parameter allows the DLMS-user to determine whether the requested DLMS was provided successfully, or failed for the reason specified. The possible value conveyed in this parameter is as follows:

- a) "success";
- b) "failure — DLM-object-identifier is unknown"; or
- c) "failure — reason unspecified".

7.6.2.3 Current-value

This parameter is present when the status parameter indicates that the requested service was performed successfully. This parameter specifies the current value for the DLM-object specified by the associated DLM-object-identifier. Its type is identical to that of the specified DLM-object.

7.7 Action

7.7.1 Function

This primitive may be used to command a specified action of the DLE.

7.7.2 Types of parameter

Table 12 lists the primitive and parameters of the ACTION DLMS.

Table 12 – DLM-ACTION primitive and parameters

DLM-ACTION Parameter name	Request	Confirm
	input	output
Desired-action	M	
Action-qualifiers	C	
Status		M
Additional-information		C

7.7.2.1 Desired-action

This parameter specifies the desired action that the DLE is to take.

7.7.2.2 Action-qualifiers

This optional parameter specifies additional action-specific parameters that serve to qualify the commanded action.

7.7.2.3 Status

This parameter allows the DLMS-user to determine whether the requested DLMS was provided successfully, or failed for the reason specified. The possible value conveyed in this parameter is as follows:

- a) "success";
- b) "failure — action is unknown";
- c) "failure — action is not permitted in current DLE state";
- d) "failure — action did not complete"; or
- e) "failure — reason unspecified".

7.7.2.4 Additional-information

This optional parameter provides action-specific additional information that augments the returned status.

7.8 Event

7.8.1 Function

This primitive is used to report the occurrence of a significant DL-event for DL-management.

7.8.2 Types of parameter

Table 13 lists the primitive and parameters of the EVENT DLMS.

Table 13 – DLM-EVENT primitive and parameters

DLM-EVENT	Indication
Parameter name	output
DLM-event-identifier	M
Additional-information	C

7.8.2.1 DLM-event-identifier

This parameter specifies the primitive or composite event within the DLE the occurrence of which is being announced. The naming-domain of the DLM-event-identifier is the DLM-local-view.

7.8.2.1.1 Additional-information

This optional parameter provides event-specific additional information.

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INTERNATIONAL
ELECTROTECHNICAL
COMMISSION

3, rue de Varembé
P.O. Box 131
CH-1211 Geneva 20
Switzerland

Tel: + 41 22 919 02 11
Fax: + 41 22 919 03 00
info@iec.ch
www.iec.ch